

From Adaptive K to Routed RTL: A CHIA Loop for HEVC RDO Co-Design

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Artifact: <https://github.com/duongnguyenhieu/CHIA-RDO>, release chia-rdo-v1.0.0

Abstract—HEVC intra coding exposes 35 luma prediction modes, making full rate-distortion optimization (RDO) expensive. Yet candidate count alone is not a hardware objective: an FPGA with P lanes executes $\lceil K/P \rceil$ batches, while replication changes area, routing, timing, and power. We propose CHIA-RDO, an evidence-preserving CHIA loop that joins pinned HM-16.20 experiments, a history-aware proposal controller, analytical screening, bit-exact Verilator regression, Pareto updates, GCP execution, and Vivado implementation. A frozen Adaptive- K policy reduces held-out RDO evaluations by 38.21% while retaining 97.91% of Full-RDO winners; its aggregate Y-PSNR BD-rate is -0.026% , interpreted as no observed penalty on this limited corpus. Resource sharing, a depth-3 quant/dequant split, corpus-qualified 31-bit cost, and a balanced reduction tree yield a routed 16-lane kernel at a derived 68.120 MHz and 8.134 M candidates/s, $2.128\times$ the P1 baseline proxy.

Index Terms—Agentic co-design, adaptive candidate selection, FPGA, HEVC, rate-distortion optimization, RTL.

I. INTRODUCTION AND BACKGROUND

HIGH Efficiency Video Coding (HEVC) improves compression by evaluating a large set of spatial prediction choices, but this flexibility makes intra-mode search computationally expensive. For each prediction unit, the HM reference encoder first applies rough mode decision (RMD), using SATD and an estimated signaling cost to rank 35 luma modes. It then performs the more expensive rate-distortion optimization (RDO), which reconstructs and compares shortlisted candidates using distortion and coded-rate estimates [3].

Fast intra-decision methods reduce this workload by pruning candidates before full RDO. Fixed-size shortlists are simple but cannot adapt to content: smooth or strongly directional blocks may need few candidates, whereas textured or ambiguous blocks benefit from broader search. Prior adaptive methods use features such as relative SATD and statistical risk to trade encoding effort for coding efficiency [2]. These methods are usually evaluated as software algorithms, without considering how each candidate budget maps to a parallel hardware scheduler.

That mapping creates a cross-layer problem. An FPGA with P lanes executes $\lceil K/P \rceil$ batches, so two policies with equal mean K can incur different latency and lane utilization. Conversely, increasing P can reduce batches while increasing area, routing pressure, and critical-path delay. Algorithm selection and RTL architecture therefore need a shared evidence loop rather than independent optimization.

CHIA-RDO addresses this gap by connecting software coding quality, cycle-accurate RTL correctness, and routed

physical feedback in one reproducible CHIA workflow. Its principal contributions are:

- 1) a composable CHIA graph connecting policy search, matched HM curves, RTL checks, cloud workers, persistent history, and backend feedback;
- 2) a frozen, independently specified Adaptive- K policy, evaluated separately from later hardware-aware policies;
- 3) a physically driven RTL evolution from monolithic replication to serial shared lanes and balanced winner reduction; and
- 4) machine-readable identities, hashes, cache behavior, and Pareto history supporting replay and claim auditing.

II. ADAPTIVE- K ALGORITHM AND CONTRACT

The policy treats a low normalized best rough cost as evidence that one mode already predicts the block well, so a short expensive-RDO prefix is sufficient; ambiguous blocks retain more candidates. For each luma PU, it:

- 1) computes $C[m] = \text{SATD}[m] + \text{bits}[m]\sqrt{\lambda}$ for every mode $m = 0, \dots, 34$;
- 2) stable-sorts modes by increasing C , retaining lower mode ID on an exact tie;
- 3) computes

$$\hat{C}_{\min} = \frac{\min_m C[m]}{WH(2^b - 1)}, \quad (1)$$

for PU size $W \times H$ and luma bit depth b ;

- 4) selects

$$K = \begin{cases} 4, & \hat{C}_{\min} \leq 0.04, \\ 16, & 0.04 < \hat{C}_{\min} \leq 0.20, \\ 35, & \hat{C}_{\min} > 0.20; \end{cases} \quad (2)$$

- 5) runs candidate RDO on exactly the first K ranked modes, retaining the earlier stable rank on an exact RD-cost tie.

Unlike our earlier confidence-gap and relative-SATD candidates, this rule uses an absolute rough-cost difficulty measure normalized across PU area and sample range. A bounded comparison on two deterministic discovery workloads selected the tuple (0.04, 0.20; 4, 16, 35) as the lowest-cycle tested adaptive point below the predeclared 1% BD-rate gate. We then froze its identity and evaluated it without retuning on three predeclared held-out synthetic families.

The controller deliberately excludes P , scheduler state, QP, and activity; hardware latency follows only afterward as

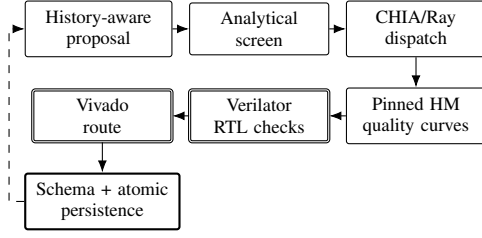


Fig. 1. The complete CHIA-RDO loop. Persisted Pareto and failed-point history guides the next proposal.

$\lceil K/P \rceil$ batches. Thus RTL exploration cannot silently alter coding decisions. All matched curves use pinned HM-16.20, Main 8-bit 4:2:0 All-Intra, four frames, and QP 22/27/32/37.

III. END-TO-END CO-DESIGN ARCHITECTURE

Native `ChiaFunction` nodes pass Ray object references asynchronously. The deterministic controller consumes completed records, excludes existing experiment IDs, and rewards novelty and Pareto-boundary exploration. The initial graph screened 72 (P , depth, buffer, width) points. A multi-wave campaign accumulated 114 hardware/history records, 107 successful logical experiments, 11 matched HM curves (132 encodes), and a 10-point Pareto set. The calibrated graph later screened 832 combinations over $P = 1 \dots 16$, depth 2/3, and cost width 31–56. It produced 16 Pareto points; deliberate replay returned 832/832 cache hits with no new evaluation.

IV. RTL ARCHITECTURE AND VERIFICATION

The hardware/software boundary is explicit. HM software evaluates rough costs for all 35 modes, creates the stable ranking, applies Adaptive- K , obtains each candidate’s integer HM bit count, quantizes λ to Q16.16, and packages the event. Adaptive- K , rough ranking, and CABAC-context bit estimation are therefore not implemented in RTL.

A start transaction supplies K ; up to 35 ranked 6-bit modes and aligned 24-bit rate counts; 6-bit QP; 32-bit λ ; seventeen 8-bit top/left references; and the row-major 4×4 8-bit original luma block. The final `full_rdo_pway_optimized` elaboration uses $P = 16$, depth 3, and a 31-bit cost. Its ready/valid scheduler dispatches candidate $bP + l$ to lane l and masks the partial final batch.

Hardware performs 4×4 luma prediction, residual, two-pass integer DST, scalar quant/dequant, inverse DST, reconstruction, SSE distortion, fixed-point RD cost, and stable winner reduction. Per-lane retirement exposes accepted rank, mode, support, and cost; completion returns winner rank/mode/cost plus cycle, batch, stall, and lane-activity counters. Lower cost wins, with stable rank breaking exact ties. A batch retires in 134 cycles without backpressure.

A. Verification Evidence

Final campaigns executed 121,600 vector/candidate checks. The depth-3 serial corpus alone contains 8,960 vectors covering all 35 modes and four QPs; the tree campaign adds 10,560 P1/P8/P16 evaluations including backpressure. Stage values, order, winners, cycle counts, dropped/duplicate checks, and interfaces all passed with zero failures.

V. CODING AND HARDWARE EVALUATION

The frozen policy responds to content. It removes 13.95% and 13.30% of RDO work on mixed-texture and repetitive families, but 87.36% on smooth/edge content. Worst individual-run winner retention is 91.00%. On the held-out aggregate, Fixed- K 16 removes more evaluations (54.29%) but retains only 85.70% of Full-RDO winners. Track-B is a separately versioned, more aggressive hardware-oriented point; it is not evidence that the frozen policy was altered after selection.

The Gwon–Choi relative-SATD Bayesian method reports 31.54% average encoding time reduction and +0.93% BD-rate against HM-16.6 under its own Intra Main condition [2]. It starts from stock-HM RMD modes and includes a minimum-risk Bayesian classifier. Our relative-SATD baseline instead starts from all 35 modes and uses threshold counting only, so the published result is context rather than a direct comparison.

VI. PHYSICAL FEEDBACK AS AN AGENT INPUT

The first P-way baseline replicated a monolithic candidate datapath. At P1, the critical path traversed prediction, residual, both forward-transform passes, quant/dequant, both inverse passes, reconstruction, and distortion: 170 logic levels, including 69 CARRY8 levels. P2 reached 175 levels, and P4 exceeded LUT/CARRY capacity before placement.

The next proposal serialized work within each lane and inserted state boundaries between major arithmetic phases. This raised Fmax and reduced per-lane area, allowing P16 to route, but a serial 16-way winner scan became an 80-level path. The final balanced tree reduced that path; the routed critical path is now 39 levels inside one lane’s prediction/residual datapath and ends at a residual-sample RAM input.

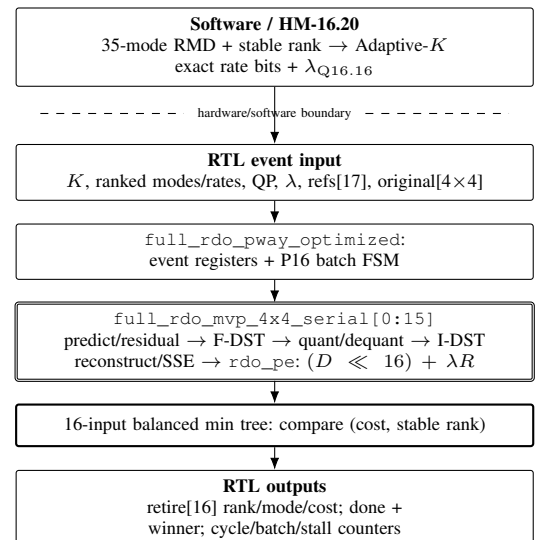


Fig. 2. Software/HM partition and the selected RTL hierarchy. Each of the 16 lanes is internally serial and takes 132 cycles per candidate.

TABLE I
CODING RESULTS RELATIVE TO THE FULL-RDO ANCHOR

Method / Corpus	Candidate Rule	Avg. K	RDO Red.	Retention	BD-Rate
Full-RDO anchor	All 35 rough-ranked modes	35.000	0.00%	100.00%	0.000%
Fixed- $K8$ / discovery	First 8 ranked modes	8.000	77.14%	68.05%	+3.574%
Adaptive- $v0$ / discovery	Confidence gap; $K = 4/8/16$	11.568	66.95%	75.67%	+2.866%
Relative-SATD / discovery	Threshold count, quantized K	25.624	26.79%	96.07%	+0.528%
Frozen threshold / discovery	Normalized rough cost; $K = 4/16/35$	18.651	46.71%	90.25%	+0.755%
Frozen threshold / held-out	Same policy, no retuning	21.628	38.21%	97.91%	-0.026%
HW-aware Track-B / P8	Joint quality/cycle objective	16.456	52.98%	88.02%	-0.054%

The held-out set contains three synthetic families; negative aggregate BD-rate is not claimed as general quality improvement.

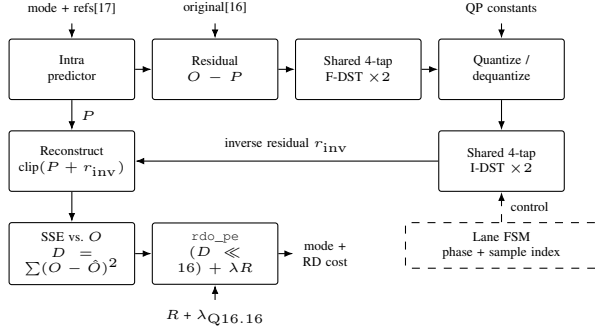


Fig. 3. Datapath inside one serial RTL lane. Registers between FSM phases store the sixteen intermediate samples; the same arithmetic resources are reused across positions. The FSM advances the ordered phases one sample or coefficient per cycle. Sixteen identical lanes feed the reduction tree in Fig. 2.

A. Measured Interpretation

To compare routed variants under a common 5 ns constraint, we report

$$f_{\text{derived}} = \frac{1}{5 \text{ ns} - \text{WNS}}, \quad T_{\text{kernel}} = \frac{P f_{\text{derived}}}{C_{\text{batch}}}. \quad (3)$$

This post-route comparison proxy is distinct from encoder frames/s. The selected P16 design routes with $\text{WNS} = -9.680$ ns, hence $f_{\text{derived}} = 68.120$ MHz and $T_{\text{kernel}} = 8.134$ M candidates/s for 134-cycle batches.

B. Resource Budget

On the KV260 device (117,120 LUTs, 1,248 DSPs), the final design uses 52.26% of LUTs and 65.38% of DSPs. Against the project’s 70%-of-device integration budget, this is 74.66% of the LUT allowance and 93.47% of the DSP allowance, leaving 57 DSPs. P16 is the throughput endpoint of the Pareto frontier, not the minimum-area or minimum-power point.

VII. ROUTED MICROARCHITECTURE COMPARISON

The balanced tree costs 975 LUTs and 45 FFs over the P16 serial scan, leaves DSP count unchanged, and reduces estimated power from 2.338 W to 2.288 W. It improves derived Fmax by 64.1% and kernel rate by 64.1%. Relative to the monolithic P1 proxy, its candidate rate is 2.128 $\times$; relative to P2 it is 1.133 $\times$. The calibrated CHIA model predicts P16 within 0.14% for LUT, 0.45% for Fmax/throughput, and 0.43% for power.

VIII. REPRODUCIBILITY AND DELIVERABLES

Local/GCP parity produced identical bitstream, reconstruction, and trace hashes for a reference encode. GCP workers later replayed RTL campaigns with zero failures. Post-run Compute Engine queries found no remaining instances, disks, or reserved addresses. Actual billed spend and promotional-credit balance remain unavailable and are not replaced by estimates.

The artifact preserves policy versions, manifests, source and corpus hashes, raw Vivado reports, machine-readable result schemas, and Pareto history. Principal deliverables are CHIA proposal/evaluation graphs and the deterministic policy controller; pinned HM patches, matched quality curves, and synthetic fixtures; parameterized SystemVerilog, golden vectors, and Verilator campaigns; and analytical replay, model calibration, routed reports, and parsers.

IX. LIMITATIONS AND NEXT STEP

The coding study uses small deterministic synthetic All-Intra workloads, not standard natural-video classes. RTL is limited to 4 $\times$ 4 8-bit luma and excludes chroma, inter prediction, larger transforms, RDOQ, syntax coding, and hardware CABAC/context evolution. Exact HM candidate bit counts are supplied externally. The 31-bit cost width is corpus-qualified, not formally bounded for arbitrary input.

Resource scale is also a limitation. The selected design uses 816 DSP48E2 primitives, exactly 51 per lane times 16 lanes. “Serial” describes candidate execution within a lane; it does not imply one shared multiplier. Runtime products in prediction, quant/dequant, squared error, and the 24 $\times$ 32-bit λR term infer multiple DSP slices, and the RTL does not explicitly share these multipliers across lanes. The P16 scan already uses 60,232 LUTs; the balanced tree adds 975 LUTs and 45 FFs but no DSPs, producing the 61,207-LUT total. A final hierarchical utilization split by operation was

TABLE II
ROUTED IMPLEMENTATIONS

Architecture	P	LUT/DSP	WNS	MHz	M/s
Mono. combinational	1	36,454/178	-60.395	15.292	3.823
Mono. combinational	2	70,817/356	-64.653	14.357	7.178
Serial d3, w31	8	30,813/408	-9.620	68.399	4.084
Serial P16 scan	16	60,232/816	-19.092	41.508	4.956
Serial P16 tree	16	61,207/816	-9.680	68.120	8.134

All rows routed. WNS is in ns; MHz is derived from the routed critical path; M/s is the candidate-kernel rate.

not preserved, so finer DSP attribution would require a new synthesis report.

All physical runs are out of context and omit board-level clocks, pins, AXI, and encoder integration. The routed result is therefore reported using its derived critical-path frequency. Future work will pipeline forward-transform accumulation and repeat regression, routing, and board-level integration.

Within these limits, CHIA-RDO demonstrates the intended agentic co-design behavior: it preserves a software-quality contract, searches reproducibly, checks a bit-exact kernel, and uses failed backend evidence to produce a specific mi-

croarchitectural revision rather than merely requesting more parallelism.

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